Capture the Flag Project

By:Team Hacksquad

Introduction

Heaven Montoya: 40 hours (challenge design, testing, advertisement/sign up)

Cristal Legarda: 40 hours (challenge design, environment set up, testing)

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Project Introduction

For our project we created a capture the flag event. It included a fully functional CTFd platform, with multiple security challenges of multiple categories that simulated real world scenarios. To set it up we used a combination of tools including MySQL, Redis, and Nginx. The goal for our project was to design, build, and host an educational capture the flag event to help students at New Mexico State University increase there cybersecurity skills.

Project Constraints

- Providing an education and beginner friendly challenges
- Time to design, build, and host event
- Strict project deadlines and presentation requirements
- Limited VM resources
- Restricted to open source tools
- Maintaining and secure and accessible environment
- Limited team availability & Scheduling

Expected and Final Deliverables

Expected Deliverables:

- Fully Functional CTFd platform
- 15+ challenges
- Working scoreboard and usr system
- Smooth event hosting were participant join easily with no issues
- Challenges that ranged in different complexity level

Final Deliverables:

- Successfully deployed a CTFd platform
- 18 challenges designed and tested
- Fully functional scoreboard and user system
- The event went successfully with no issues
- Consisted of each category having challenges of different complexity levels

Timeline

Week 1:

- Agreed on project goals
- Request server
- Set up VM

Week 2:

- Started configuring CTFd environment
- Began researching challenge ideas & tools

Week 3:

- Finished Configuring CTFd environment
- Began creating challenge drafts

Week 4:

- Began creating Challenges

Week 5:

- Continued creating challenges

Week 6:

- Continued creating challenges
- Started testing our challenges

Week 7:

- Started testing the entire environment
- And finished challenges
- Advertised Event

Week 8:

- Added final touches
- Hosted event

Testing/Validation of final deliverables

Platform Testing

- Verified CTFd installation and login
- Confirmed users registration, and scoreboard worked correctly
- Tested the platforms updates in real time

Challenge Testing:

- Loaded properly
- Had the correct flag
- Challenge was solvable with intended method
- Ensured no shortcuts

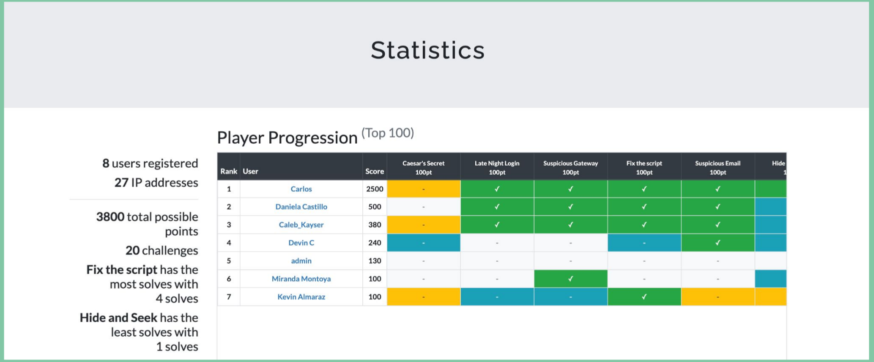
Performance & Reliability Testing:

- Ensured CTFd platform stability under load
- Tested CTFd and Docker containers
- Confirmed the platform was fully functional automatically after VM reboots.

User Experience:

- Ran through the system with a test user
- Checked if challenges were clear and accurate
- Checked if flags were correct
- Ensured nothing was broken

Visuals



Statistics

8 users registered
27 IP addresses

3800 total possible points

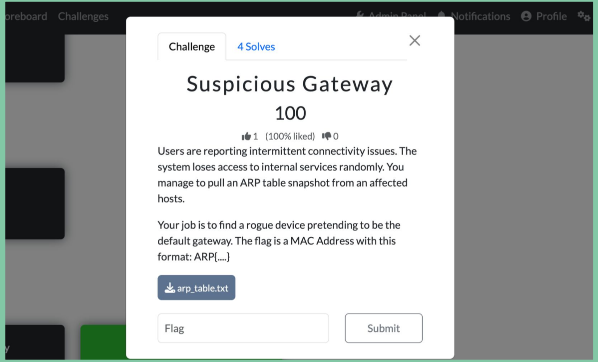
20 challenges

Fix the script has the most solves with 4 solves

Hide and Seek has the least solves with 1 solves

Player Progression (Top 100)

Rank	User	Score	Caesar's Secret 100pt	Late Night Login 100pt	Suspicious Gateway 100pt	Fix the script 100pt	Suspicious Email 100pt	Hide
1	Carlos	2500	-	✓	✓	✓	✓	
2	Daniela Castillo	500	-	✓	✓	✓	✓	
3	Caleb_Kayser	380	-	✓	✓	✓	✓	
4	Devin C	240	-	-	-	-	✓	
5	admin	130	-	-	-	-	-	
6	Miranda Montoya	100	-	-	✓	-	-	
7	Kevin Almaraz	100	-	-	-	✓	-	



nmsuctf.org

Username- admin

Password - root

Key Challenges

Limited VM resources:

Low CPU and Ram caused slow performance and could only allow one team member to work on it at once

How we overcame it:

- Reduced unnecessary services
- Scheduled different working time

Strict Deadline:

We had two months to plan, build, test, and host entire CTF event.

How we overcame it:

- Prioritize project
- Write down what we have to do in order
- Worked on project independently

Team Availability/Team Size

Team collaboration was difficult due to having different schedules along with having a small team of 3.

How we overcame it:

- Divided project task
- Used asynchronous communication

Overview

To finish, our project involved creating and hosting a fun and educational event. During the presentation, we talked about the goals of the project, the challenges and constraints we faced, the work we did to build and test everything, and how we made sure the event worked smoothly. Overall, this project helped us learn new skills, solve problems, and work together as a team to create a fun event for ICT students.

Thank you for listening!